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Lecture 3

Linear Regression

Hauhechel-Bläuling (*Polyommatus icarus*)

- ♦ Der Hauhechel-Bläuling ist ein Schmetterling (Tagfalter) aus der Familie der Bläulinge (Lycaenidae). Es gibt verschiedenste Bläulingsarten, und wie der Name schon sagt, weisen sie sich durch eine bläuliche Färbung aus. Der Hauhechel-Bläuling ist der häufigste.
- ♦ Der Hauhechel-Bläuling hat eine Flügelspannweite von 25 bis 30 Millimetern.
- ♦ Die Männchen sind auf der Flügeloberseite intensiv blau gefärbt. Weibchen hingegen sind vor allem braun, haben aber dennoch einen Blaustich. Die Flügelunterseite ist auf hellem Grund auffällig schwarz und orange gepunktet und die Flügel selbst sind gesäumt von einer hellen Fransenborte.

Quellen:

* <https://de.wikipedia.org/wiki/Hauhechel-Bläuling>

* <https://www.plantura.garden/gruenes-leben/schmetterlinge-die-10-beliebtesten-heimischen-schmetterlingsarten>



Photo by kie-ker on pixabay



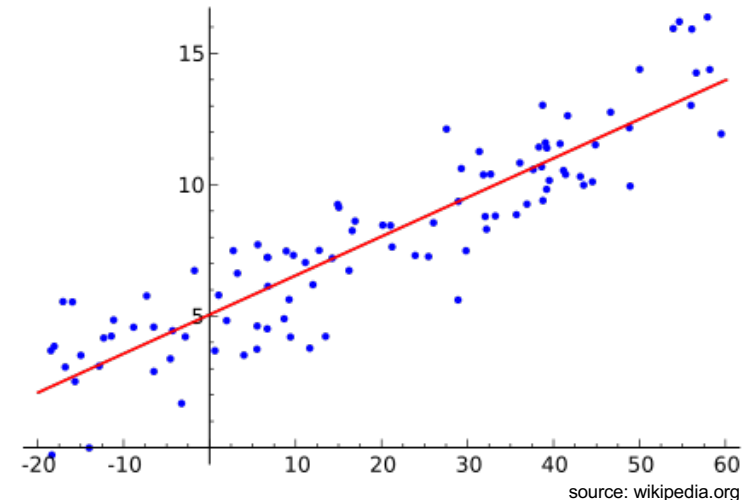
Lecture – Linear Regression

1. Motivation
2. Simple Linear Regression
3. Regression Model Evaluation
4. Multiple Regression



What is Regression Analysis?

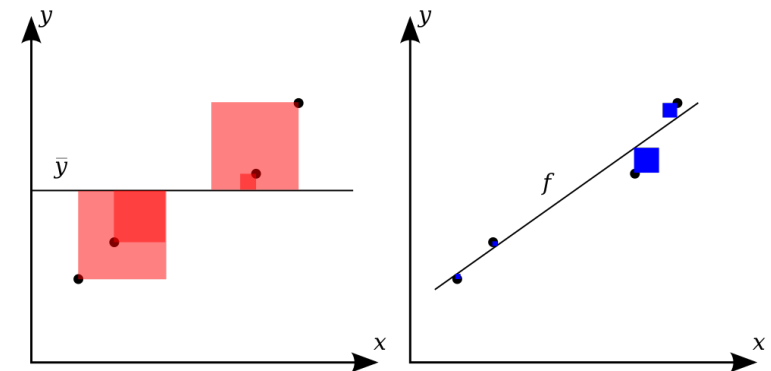
- ◆ **Regression Analysis:** statistical process for estimating the relationships among variables
 - relationship between a **dependent variable** & one or more **independent variables** ('predictors')
 - how does the value of the dependent variable change when one of the independent variables is varied (while the other independent variables are held fixed)
- ◆ **Regression Function:** function of the independent variables
- ◆ **Usage:**
 - prediction/forecasting of dependent variable given the independent variables
 - quantify strength of the relationship between the dependent variable and the independent variables





Regression Models

- ♦ **Regression models** relate Y to a function of X and β : $Y \approx f(X, \beta)$ with
 - the dependent variable Y ,
 - the independent variables X ,
 - the unknown *parameters* β , a vector of size k . (also called *regression coefficients*)
- ♦ Form of the function f must be specified → e.g. linear regression
- ♦ Typically, n data points of the form (Y, X) are observed ($n > k$)
 - enough information to estimate a unique value for β that best fits the data in some sense
 - regression model can be viewed as an overdetermined system in β .
 - regression analysis finds a solution for unknown parameters β that will e.g. minimize the distance between the measured and predicted values of the dependent variable Y (=method of least squares).



source: wikipedia.org



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Simple Linear Regression Model

- ♦ **Simple**: Y and X are scalar values $\rightarrow n$ data points of the form (y_i, x_i) are observed

- ♦ **Linear**: linear relationship between dependent and independent variables:

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$$

- ♦ ε_i : stochastic **error term** (also called *disturbance term*, or *noise*): captures all factors influencing the dependent variable y_i other than x_i

- ♦ Graphically: draw a line through the data points

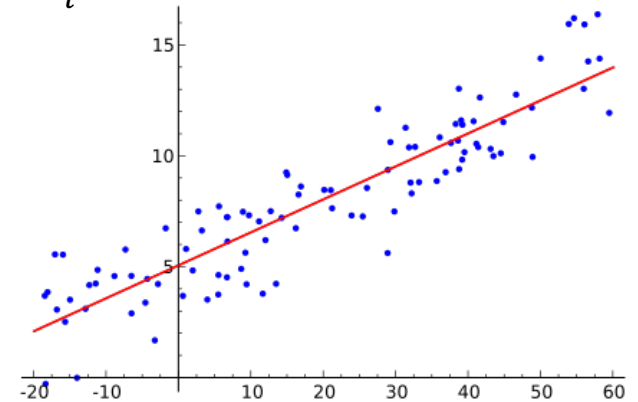
$$f(x) = \beta_0 + \beta_1 x$$

β_0 = y-intercept („Achsenabschnitt“)

β_1 = slope („Steigung“):

- ♦ Assumption (commonly)

ε_i follow the gaussian distribution with mean 0 and identical variance: $\varepsilon_i \sim \mathcal{N}(0, \sigma^2)$



source: wikipedia.org



Estimating the regression coefficients (parameter vector)

Ordinary least squares (OLS)

- ◆ simplest (and most common) estimator
- ◆ minimize the sum of squared errors

$$\operatorname{argmin}_{\beta_0, \beta_1} \sum_{i=1}^n \varepsilon_i^2 = \operatorname{argmin}_{\beta_0, \beta_1} \sum_{i=1}^n (y_i - (\beta_0 + \beta_1 x_i))^2$$

- ◆ computing the partial derivatives and setting them to 0 leads to

$$b_1 = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2} \quad \text{and} \quad b_0 = \bar{y} - b_1 \bar{x}$$

where b_0 is the estimation for β_0 and b_1 for β_1 .

- ◆ Many other estimators exist (especially important if the assumptions regarding the error are not met).



Lecture – Linear Regression

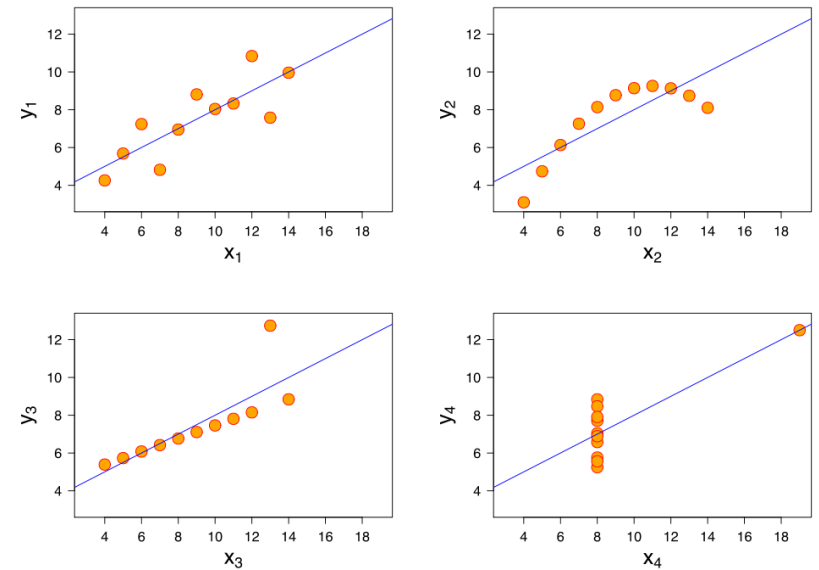
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Why Do We Need to Evaluate Regression Models?

- ◆ Different data sets can lead to the same linear regression model, e.g.
https://en.wikipedia.org/wiki/Anscombe%27s_quartet
- ◆ model is only useful if we can assess its performance

➔ Measure for the *goodness of the prediction* needed



Anscombe's quartet: four datasets that have nearly identical simple descriptive statistics

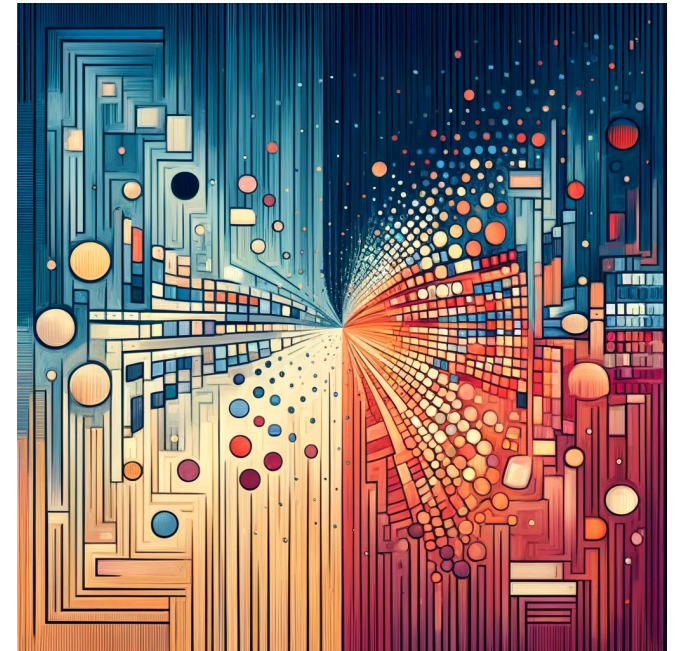
source: wikipedia.org

- ◆ Different use cases require different evaluation metrics
- ◆ A good fit does not always mean a good model



Classification Evaluation to Regression Evaluation

- ◆ How Is Regression Evaluation Different from Classification?
 - **Classification**: Accuracy, Precision, Recall, F1-score
 - **Regression**: We can't just say "right" or "wrong"
 - ➔ Measure the difference between predicted and actual values
- ◆ Types of Regression Evaluation Metrics
 - **Error-based metrics**
Measure how much the predictions deviate from actual values
 - e.g., **MSE**, **RMSE**, **MAE**
 - **Goodness-of-fit metrics**
Measure how well the model explains variability
 - e.g., **R^2** or **Adjusted R^2**
 - **Relative performance metrics**
Evaluate error relative to scale
 - e.g., **MAPE**, **MSLE**





What Makes a Good Regression Metric?

- ◆ No single metric is perfect—each has trade-offs!
- ◆ Considerations when choosing a metric:
 - **Interpretability**: Can stakeholders understand the metric?
 - **Sensitivity to outliers**: Do we want to penalize large errors more?
 - **Scale dependence**: Does the metric depend on the units of measurement?
 - **Comparability**: Can we compare models across datasets?
 - **Business problem**: Does the metric fulfil the requirements of the application scenario?
- ◆ Frequently: evaluate regression model using multiple metrics together



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 - 3.3 Relative performance metrics
 - 3.4 Evaluation metrics and loss function
4. Multiple Regression



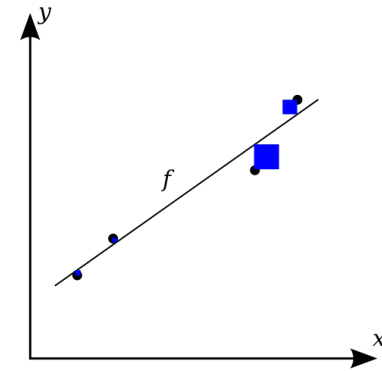
What Are Error-Based Metrics?

- ◆ Measure how far predictions are from actual values
- ◆ Lower values indicate a better model
 - Note: scikit-learn convention: higher values indicate better models
→ use “negative” error-based metrics
- ◆ Different metrics penalize errors differently
- ◆ Most common metrics
 - **MSE** mean squared error
 - **RMSE** root mean squared error
 - **MAE** mean absolute error



MSE = Mean Squared Error

$$MSE = \frac{1}{n} \sum_i (y_i - f(x_i))^2$$



- ◆ Measures average squared difference between predictions and actual values
- ◆ Squaring errors amplifies large mistakes
- ◆ Widely used in optimization (OLS regression minimizes MSE)



RMSE = Root Mean Squared Error

$$RMSE = \sqrt{MSE}$$

- ◆ Same unit as the target variable (easier to interpret)
- ◆ Still penalizes large errors but easier to understand than MSE
- ◆ Commonly used in real-world applications



MAE = Mean Absolute Error

$$MAE = \frac{1}{n} \sum_i |y_i - f(x_i)|$$

- ◆ Measures absolute differences instead of squared differences
- ◆ Less sensitive to outliers compared to MSE
- ◆ More interpretable but not differentiable at zero, which affects optimization



Comparing MSE, RMSE, and MAE

- ◆ MSE
 - + Penalizes large errors more
 - Hard to interpret (squared units)
- ◆ RMSE
 - + Same unit as target variable
 - Still sensitive to outliers
- ◆ MAE
 - + Easier to interpret
 - Doesn't penalize large errors strongly
- ◆ MSE & RMSE emphasize large errors more (good when large mistakes are critical)
- ◆ MAE is more robust to outliers but treats all errors equally



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What Are Goodness-of-Fit Metrics?

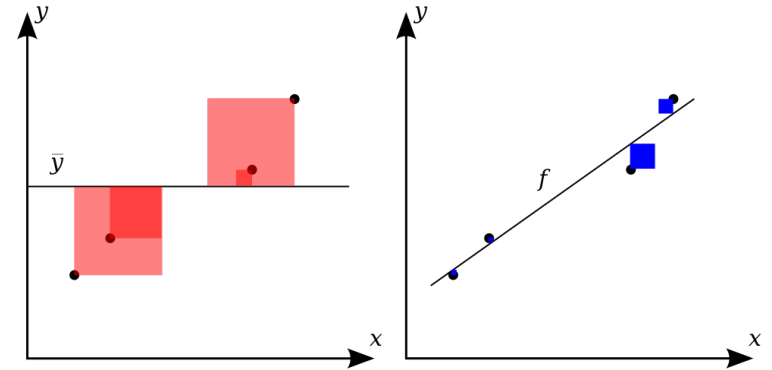
- ◆ Measure how well the model explains variability in the target variable.
- ◆ Higher values indicate a better model
- ◆ Help compare models beyond just error-based scores
- ◆ Most common metrics
 - R^2
 - Adjusted R^2



R^2 = Coefficient of Determination

1- is used to norm the result, higher value means better model

$$R^2 = 1 - \frac{\sum_i (y_i - f(x_i))^2}{\sum_i (y_i - \bar{y})^2}$$



- ◆ Measures the proportion of variance in y explained by the model.
 - $R^2 = 1 \rightarrow$ Perfect model (all variance explained)
 - $R^2 = 0 \rightarrow$ Model is as bad as just predicting the mean of y
 - $R^2 < 0 \rightarrow$ Model is worse than using the mean as a prediction
- ◆ Note: sometimes lower case is used: R^2 or r^2



Limitations of R^2

- ♦ Adding more features (almost) always increases R^2 (even if they don't help)
 - ♦ A high R^2 does not mean the model is good—it could be overfitting
 - ♦ Does not tell us if the model is useful in practice
 - ♦ Never use R^2 to compare models!
- R^2 is a “dangerous” metrics
looks easy, but very hard to understand the details and use correctly!



Adjusted R^2

$$\text{Adjusted } R^2 = 1 - (1 - R^2) \times \frac{n-1}{n-p-1}$$

n = number of observations

p = number of predictors

- ◆ Adjusts for the number of predictors: Penalizes adding unnecessary predictors
- ◆ Preferred for comparing models with different numbers of predictors
(→ never look at R^2 when adding a feature, always look at Adjusted R^2)
- ◆ But still does not indicate whether
 - the independent variables are a cause of the changes in the dependent variable
 - the correct regression was used – the model is overfitting or underfitting
 - the most appropriate set of independent variables has been chosen
 - ...



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What Are Relative Performance Metrics?

- ◆ Absolute errors (MSE, RMSE, MAE) depend on the scale of the data
 - Problem: A \$10 error is huge for predicting coffee prices but tiny for real estate prices
- ◆ Relative Performance Metrics: Measure error in relation to the true value
- ◆ Lower values indicate a better model
- ◆ Most common metrics
 - **MAPE** mean absolute percentage error
 - **MSLE** mean squared logarithmic error



MAPE = Mean Absolute Percentage Error

$$\text{MAPE} = \frac{1}{n} \sum_i \left| \frac{y_i - f(x_i)}{y_i} \right|$$

n = number of observations

- ◆ Measures error as a percentage of the actual value
- ◆ Pros
 - Easy to interpret (percentage-based)
 - Useful when comparing across different scales
- ◆ Cons
 - Undefined if $y_i = 0$
 - Over-penalizes errors for small values



MSLE = Mean Squared Logarithmic Error

$$\text{MSLE} = \frac{1}{n} \sum_i (\log(1 + y_i) - \log(1 + f(x_i)))^2$$

n = number of observations

- ◆ Key idea: Penalizes relative differences instead of absolute ones.
- ◆ Pros
 - Useful for data with exponential growth (e.g., population, revenue, stock prices)
 - Useful for situations where under-predictions matter more than over-predictions
- ◆ Cons
 - Not intuitive



Comparing MAPE and MSLE

- ◆ MAPE

- + Easy to interpret (percent-based)
- Breaks for zeros, over-penalizes small values

- ◆ MSLE

- + Works for exponential growth, avoids over-penalizing large values
- Not intuitive, penalizes under-predictions more



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Evaluation metrics and loss function

- ◆ Pick **one loss function**
 - Should be differentiable
 - Take business application into account
 - Take data into account (scales, outliers, ...)
 - Popular choice: MSE (but sensitive to outliers)
 - Loss function = Hyperparameter – can be tuned!

- ◆ Apply **multiple evaluation metrics** to the resulting model
 - Evaluate your model from different perspectives
 - Take business application into account
 - Take stakeholders into account
 - Popular choices: RMSE, MAE, MAPE

- ◆ More advanced loss functions exist
 - e.g., Huber loss: “differentiable combination of MSE and MAE” (→ less sensitive to outliers)



Exercises

Exercise 1 – The Chocolate Bar





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Multiple Linear Regression

- ♦ **Multiple**: the scalar variable y is dependent on multiple values $x_{i1}, x_{i2}, \dots, x_{ik}$:

$$y_i = \beta_0 + x_{i1}\beta_1 + x_{i2}\beta_2 + \dots + x_{ik}\beta_k + \varepsilon_i$$

leading (for n data points) to the following equations:

$$y_1 = \beta_0 + x_{11}\beta_1 + x_{12}\beta_2 + \dots + x_{1k}\beta_k + \varepsilon_1$$

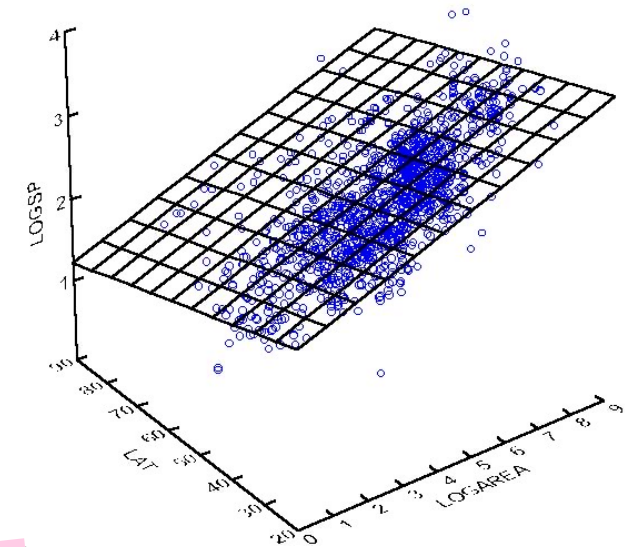
$$y_2 = \beta_0 + x_{21}\beta_1 + x_{22}\beta_2 + \dots + x_{2k}\beta_k + \varepsilon_2$$

...

$$y_n = \beta_0 + x_{n1}\beta_1 + x_{n2}\beta_2 + \dots + x_{nk}\beta_k + \varepsilon_n$$

which can be written more concisely as $\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$ with

$$\mathbf{y} = \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix} \quad \boldsymbol{\varepsilon} = \begin{pmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \vdots \\ \varepsilon_n \end{pmatrix} \quad \boldsymbol{\beta} = \begin{pmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_k \end{pmatrix} \quad \text{and} \quad \mathbf{X} = \begin{pmatrix} 1 & x_{11} & x_{12} & \dots & x_{1k} \\ 1 & x_{21} & x_{22} & \dots & x_{2k} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_{n1} & x_{n2} & \dots & x_{nk} \end{pmatrix}$$



source: <http://ordination.okstate.edu/MULTIPLE.htm>



Estimating the regression coefficients for multiple linear regression

- ◆ **OLS** can be used in basically the same way to minimize the Euclidian norm $\|y - X\beta\|_2$
- ◆ Setting the partial derivatives to 0 and solving for the regression coefficients $\mathbf{b}$ as an estimation for β leads to

$$\mathbf{b} = \begin{pmatrix} b_0 \\ b_1 \\ \vdots \\ b_k \end{pmatrix} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y}$$

- ◆ Note: Computing the inverse matrix is numerically instable (and often not possible). We do not directly do this in practise! Instead, we use SVD (singular value decomposition) to compute the pseudo inverse, which is always possible and numerically stable. Details in the Literature.



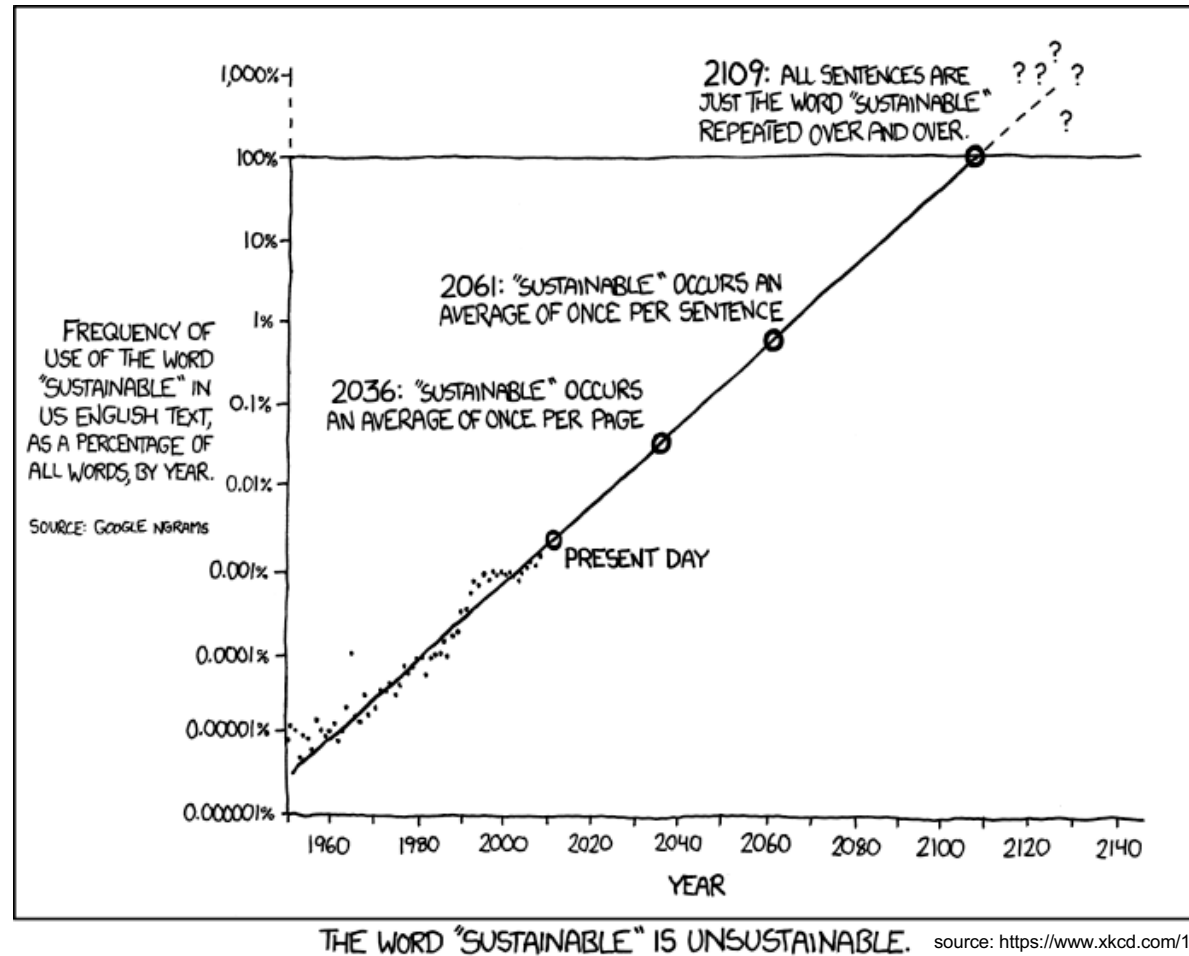
Exercises

Exercise 2 – The Cheese





The dangers of using linear regression for extrapolation

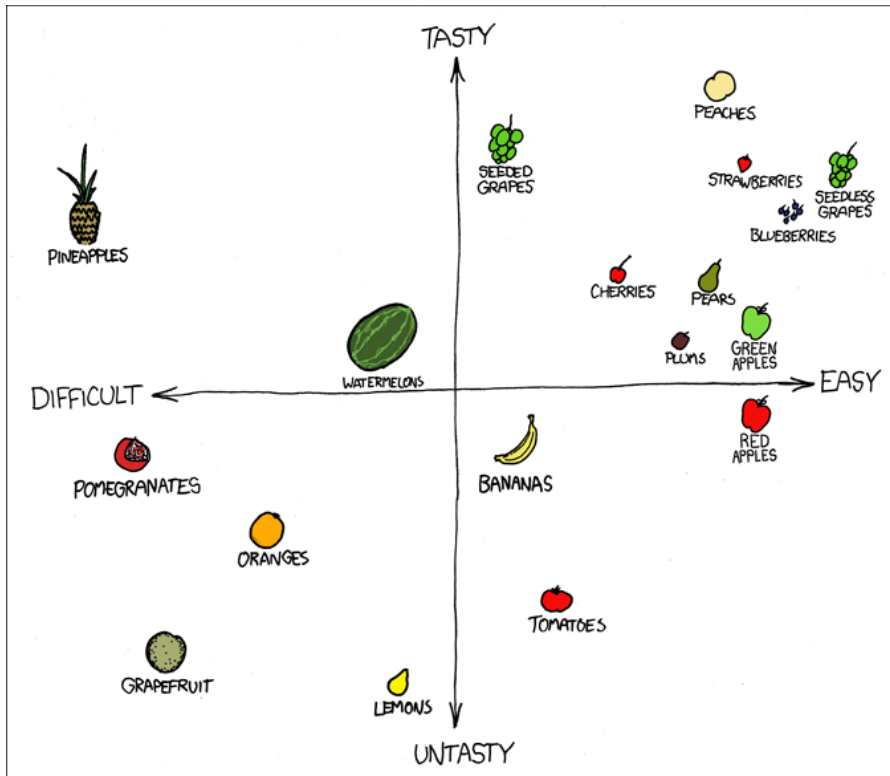




Key Takeaways

- ◆ Regression
- ◆ Linear Regression (Simple and Multiple)
- ◆ OLS – Ordinary Least Squares Optimization
- ◆ Coefficient of determination (r^2) and SSE, MSE, RMSE





source: <https://xkcd.com/388/>

Further Reading:

https://de.wikipedia.org/wiki/Lineare_Regression

https://en.wikipedia.org/wiki/Linear_regression

http://scikit-learn.org/stable/modules/linear_model.html

<http://scikit-learn.org/stable/modules/generated/sklearn.preprocessing.PolynomialFeatures.html>

<https://www.youtube.com/watch?v=NJCKd98lKEQ>

<http://onlinestatbook.com/2/regression/regression.html>